# INPhO 2002

1. **Falling Raindrop** **[12]**

   A raindrop, whose initial size may be considered negligible, falls from a height $h$ and increases its size by accreting (gathering) all the mist it encounters. If the mist is of uniform density $\rho$ and extends to the ground, determine the velocity and displacement of the raindrop as a function of time? [Hint: It may be helpful to first obtain the relation between the radius of the drop and the vertical distance it has traversed.]

2. **Buoyancy and the Law of Atmospheres** **[14]**

   (a) It is known that the density $\rho$ of air decreases with height as

   $$\rho = \rho_0 e^{-\frac{y}{y_0}}$$

   where $\rho_0 = 1.25$ kg m$^{-3}$ is the density at sea level and $y_0$ is a constant. This density variation is called the law of atmospheres. Obtain this law and an estimate of $y_0$ (in meters) assuming that both the temperature of the atmosphere and the value of $g$ remain constant with height.

   (b) A large He balloon of volume 1425 m$^3$ is used to lift a payload of negligible volume and mass 400 kg. Assume that the balloon maintains constant volume as it rises.

   (i) What is the height at which the balloon is in equilibrium?

   (ii) What is the maximum height to which it rises?

   For part (b) take density of He gas $\rho_{\mathrm{He}} = 0.18$ kg m$^{-3}$ and $y_0 = 8000$ m. Neglect air resistance.

3. **A Leaky Vessel** **[11]**

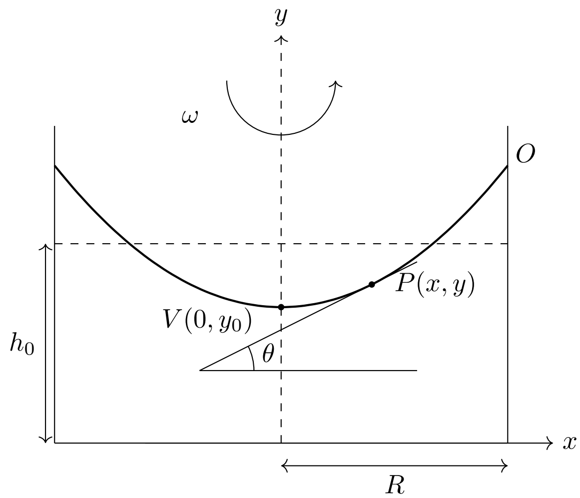
   A metallic vessel of volume $V$ with a narrow outlet tube is immersed in a water bath. The vessel contains $n$ moles of a gas at a high pressure $P$. The gas is allowed to leak out slowly to the atmosphere (pressure $P_{\mathrm{atm}}$) via the tube. The vessel and the bath are maintained at constant temperature equal to the atmospheric temperature throughout the process by dissipating heat through an electrical resistor placed inside the bath. The resistance of the resistor is $R$ and it carries current $I$. Equilibrium is attained in time $\tau$.

   (a) Obtain an expression for the change in the internal energy of the $n$ moles of the gas from the initial to the final equilibrium state. Take the molar volume of the gas at atmospheric pressure to be $v_A$.

   (b) Estimate the time $\tau$ for helium gas, which is to be considered ideal. Take $n = 5$ moles, $V = 12$ litres, $I = 1$ A, $R = 10\ \Omega$, and assume standard temperature and pressure conditions for the atmosphere.

4. **An Unusual Mirror** **[13]**

   A cylindrical container filled with mercury (Hg) rotates about a vertical axis with uniform angular velocity $\omega$. The liquid surface is curved. The figure shows a cross sectional view of the curved surface.




(a) At equilibrium show that the tangent to the surface at $P(x, y)$ makes an angle $\theta$ with the horizontal such that (see figure)

$$\tan\theta = \frac{\omega^2 x}{g} \qquad \text{(for } |x| \leq R\text{)}$$

where $R$ is the container radius.

(b) Obtain $\omega$ if the focal length of the shiny mirror surface is 20 cm.

(c) Let the angular speed given to the rotating liquid be realized by a constant power source. The rotation is started from rest at $t = 0$, then the focal length $f$ depends on time $t$ as $f \propto t^n$. Obtain the exponent $n$.

[Note: Neglect surface tension. Assume $|x| \ll R$ for part (b). Assume equilibrium is attained so that the entire mass of the liquid is rotating with the same angular speed $\omega$.]

5. **Modified Young's Apparatus** **[14]**

The Young's double-slit interference arrangement is modified by placing an isotropic transparent plate of high melting point in front of one of the slits. The refractive index of the plate is $\mu_0 = 1.5$ and its interposing thickness is $l_0 = 2$ cm. The temperature coefficient of refractive index of the plate (i.e. the fractional change in refractive index per unit rise in temperature) is $2 \times 10^{-6}/°\mathrm{C}$ and its coefficient of linear expansion $\alpha = 4 \times 10^{-5}/°\mathrm{C}$. The arrangement is illuminated by a monochromatic coherent source of wavelength $\lambda = 4600$ Å.

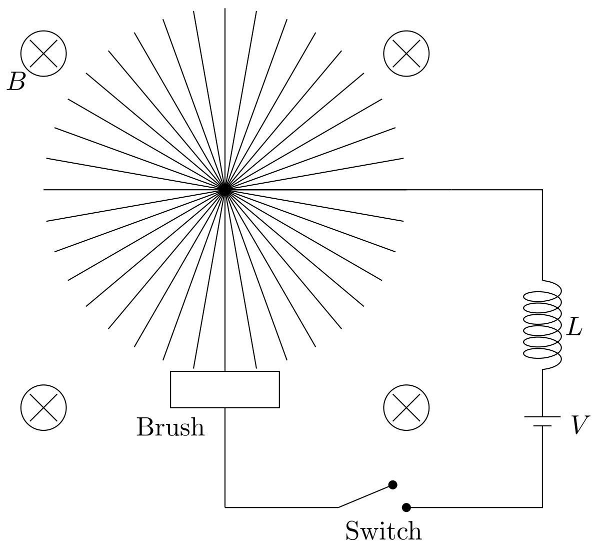
(a) The central maximum normally occurs at a spot on the screen midway between the two slits. Will this spot continue to be a maximum on insertion of the transparent plate?

(b) We confine our attention to a maximum on the screen. The plate is heated at the rate of 1°C/min, for 5 minutes. What is the fringe shift observed in this time interval?

(c) Let $I_0$ be the intensity of a maximum before the insertion of the plate. If the plate reduces the intensity of the beam through one of the slits by half, what is the intensity of the maximum in terms of $I_0$?

6. **Electromagnetic Induction** **[15]**

As shown in the figure below, a wheel consisting of a large number of thin conducting spokes is free to rotate about an axle. A brush always makes electrical contact with one spoke at a time at the bottom of the wheel. (The circuit never breaks at the brush) A battery with a voltage $V$ feeds current via an inductor of inductance $L$, into the axle, through a spoke, to the brush. A uniform magnetic field $\vec{B}$ exists perpendicular to the plane of rotation of the wheel and into the plane of the paper as shown in the figure. The radius and the moment of inertia of the wheel are $R$ and $J$ respectively. At time $t = 0$, the switch is closed. Calculate the current in the circuit and angular speed of the wheel as functions of time. Neglect friction, resistivity and gravity.




7. **The Spectra of a Star** **[10]**

Observations on a star yield a continuous spectrum with a maximum in the spectral density $u(\lambda, T)$ at $\lambda_m = 3000$ Å. One also observes discrete dark lines in the continuous spectrum, which fit the empirical formula

$$\frac{1}{\lambda} = R\left[4 - \frac{4}{n^2}\right]$$

where $n$ is an integer and the Rydberg constant $R$ has the value $1.1 \times 10^7$ m$^{-1}$. Based on the continuous spectrum estimate:

(a) The surface temperature of the star.

(b) The radiant power per unit area of the star.

Based on the discrete spectra:

(a) Deduce the atomic species responsible for the discrete spectra.

(b) Obtain the energy of emission in eV when an excited atom of this species ($n = 2$) transits to its ground state.

8. **The Magnetar** **[13]**

The magnetar is a special word coined for a neutron star with a magnetic dipole moment. The X-ray pulsar SGR-1806-20, which has been observed recently in 1998, is an example of a magnetar. Let the mass of the magnetar be the same as that of a standard neutron star, i.e. $2 \times 10^{30}$ kg and its radius be 10 km. The magnetar is spinning with a period $T = 7.5$ s and a deacceleration ('spindown') of $dT/dt = 8.0 \times 10^{-11}$. Assume that (i) the magnetic dipole moment $m$ of the magnetar is perpendicular to the angular velocity vector $\omega$ and (ii) the size of the dipole associated with the magnetar is small compared to its radius. Due to the rotation of the dipole there is a radiative energy loss whose rate $P_r$ you may take to be

$$P_r = \frac{\mu_0 m^2 \omega^4}{12\pi c^3}$$

where $c$ is the speed of light in a vacuum.

(a) What is the ratio of neutron star density to the nuclear density? [Order of magnitude answer will be sufficient.]

(b) Obtain an expression for the dipole moment of the magnetar. Numerically evaluate this dipole moment.

(c) Obtain the magnitude of the peak magnetic field $B_p$ on the surface of the magnetar.

(d) How does $B_p$ compare with the maximum magnetic field we can generate in the best of our laboratories?

9. **Field and Photo-ionization** **[18]**

The ionization of an atom by a steady electric field is called field ionization. A similar process induced by light (i.e. electromagnetic) waves is called photo-ionization.

(a) Consider a hydrogenic atom at rest in a uniform steady electric field $F_0\hat{k}$. Write down the expression for the potential energy of the electron in this field.

(b) Sketch the potential energy due to this field between $-1 < z < 1$. To make this plot select a system of units where $e = 1, 4\pi\varepsilon_0 = 1$, and $F_0 = 20$.

(c) Let the energy of the electron confined in the atom be $E$. At what field strength would the atom ionize?

(d) If we take $E = -13.6$ eV, will the numerical result for the ionizing field strength be physically acceptable?

A Rydberg hydrogenic atom is one in which the electron possesses very large quantum number, e.g. $n \approx 100$. This electron may be considered to be unbound and free for all practical purposes. Supposing such Rydberg atoms are injected uniformly into an oscillating electric field $F_0 \cos(\omega t)\hat{k}$. Note that such a field can be provided by electromagnetic wave. Let the speed of the electron at the time of injection ($t = 0$), be $v = 0$.

(e) Obtain an expression for the speed of the electron at a later time $t$.

(f) Obtain an expression for the average kinetic energy of the electron.

(g) The binding energy $E_b$ of the Rydberg electron may be taken as $10^{-3}$ eV. Assuming that the criterion for photo-ionization is that the average kinetic energy exceed $E_b$, obtain the value of the ionizing field strength for microwave radiation of angular frequency $10^{10}$ rad s$^{-1}$.